

What Does a (Lensed) Galaxy Look Like?

Reproducing and Validating Score-Based Posterior Reconstruction
in Strong Gravitational Lensing

Yaocong (Derrick) Deng

MPhil in Data Intensive Science | University of Cambridge

Executive takeaway. This project independently reproduced the principal fixed-lens demonstrations of score-based source reconstruction. Posterior samples recovered the dominant source morphology, and a qualitative noise experiment reproduced the expected shift from likelihood-driven to prior-dominated inference as the observation became less informative. Additional validation revealed source-dependent residual performance and a discrepancy between generated and selected galaxy distributions, showing why visual agreement alone is insufficient to establish statistical reliability.

Motivation & Scientific Justification

Strong gravitational lensing occurs when the gravity of a foreground galaxy distorts and magnifies light from a more distant source. The resulting arcs and multiple images provide information about distant galaxies and the distribution of luminous and dark matter [1]. Future surveys are expected to discover substantially larger samples of strong lenses, creating a need for reconstruction methods that are both scalable and statistically reliable.

Recovering the unlensed source is important because assumptions about its morphology can influence estimates of the foreground mass distribution. Reliable source uncertainty therefore affects not only image reconstruction but also the scientific conclusions drawn from a lens system. This study holds the lens model fixed, allowing the reliability of source inference to be examined in isolation.

Figure 1 summarises the inference problem. A source image x is mapped by the lensing operator A to an observed image y , and reconstruction seeks the posterior distribution $p(x | y)$. The problem is ill posed because noise, finite resolution, and uneven magnification allow several source configurations to explain the same observation. A learned diffusion prior can represent a broad range of galaxy morphologies and support a distribution of plausible reconstructions rather than a single estimated image [2,3].

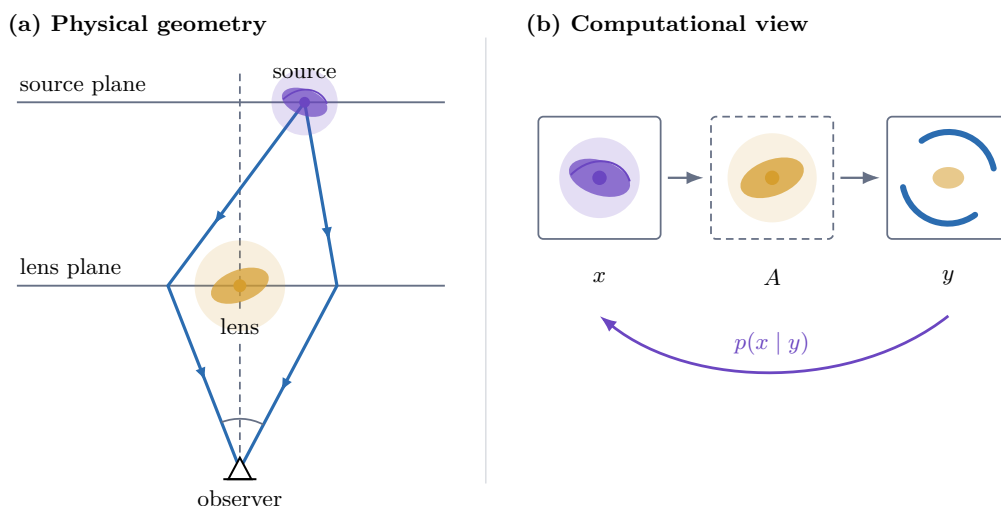


Figure 1: Physical and computational views of strong lensing. The forward model maps a source x through the fixed lens operator A to an observation y ; reconstruction targets the posterior $p(x | y)$.

Methodology

The principal fixed-lens reconstruction pipeline of Adam et al. was reimplemented independently [3]. The pipeline trains a score-based generative model on preprocessed g -band galaxy images from the PROBES compilation [4]. A fixed lens model maps known source images into the image plane, where Gaussian noise produces simulated observations. The posterior sampler then combines the learned galaxy prior with the observation likelihood to generate approximate source samples. Holding the lens parameters fixed isolates source reconstruction from uncertainty in the foreground mass model.

The reproduction covered unconditional galaxy generation, posterior source reconstruction, image-plane residual analysis, and an out-of-distribution (OOD) noise experiment. Standard reconstructions used $\sigma_N = 0.01$, 8,000 Euler–Maruyama steps, and 160 retained samples per source. The extended validation included 16 completed source runs. PQMass compared generated images with selected PROBES galaxies [5], while MIRA assessed the position of each known source within its corresponding posterior sample set [6].

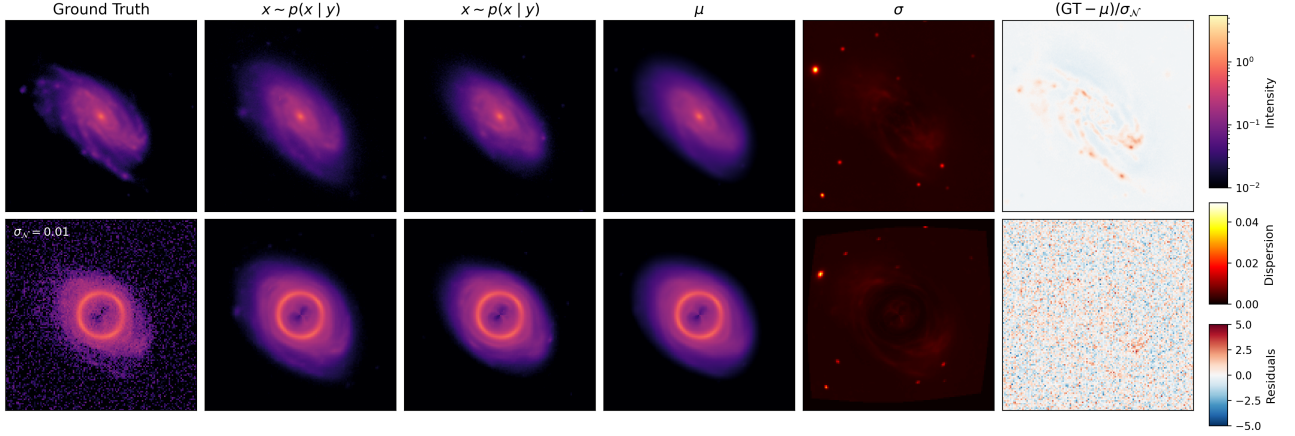


Figure 2: Complete posterior reconstruction for one simulated galaxy. The figure compares the truth, posterior draws, posterior mean, uncertainty, and source- and image-plane residuals.

Key Findings

The principal demonstrations were reproduced. Unconditional samples exhibited broad galaxy-like morphologies, although they were generally smoother and contained less resolved small-scale structure than selected PROBES images. As shown in Figure 2, posterior draws recovered the source orientation, central concentration, and dominant spiral structure. When mapped into the image plane, they reproduced the main Einstein-ring features and produced predominantly noise-like residuals.

Figure 3 illustrates how the balance between likelihood and prior changes with observation quality. At low noise, the digit-shaped source remains recognisable because the likelihood provides strong constraints. As the noise increases, the digit morphology becomes less distinct and the reconstruction shifts towards galaxy-like structures favoured by the learned prior. This experiment reproduces the expected qualitative transition, but is not intended as a quantitative performance test.

The statistical extensions placed these visual results in context. Across 16 sources, the median source-level χ^2/N was **1.058**, indicating that most reconstructions were close to the expected noise level, although performance varied across sources. PQMass detected a discrepancy between the generated and selected PROBES distributions: the mean statistic was **3.481** in pixel space and **1.695** after PCA, compared with **1.021** for the real-versus-real control.

The reconstructed posterior received a MIRA score of **0.6379**, compared with a finite-sample matched reference of **0.66875**. This difference was smaller than the Monte Carlo and bootstrap uncertainty estimates, leaving the result compatible with the matched reference at the available precision. The calibration assessment was limited by the small number of distinct truths and by the shared observation-noise realisation and aligned sampling streams, which do not represent independent repetitions of the full simulation procedure. No matched conventional baseline was included, so the study does not establish a comparative advantage over simpler reconstruction methods.

Recommendations & Next Steps

Posterior validation should be expanded using more independently regenerated truth–observation pairs, each with its own noise realisation and posterior sampling stream. This would increase the number of independent experimental units more

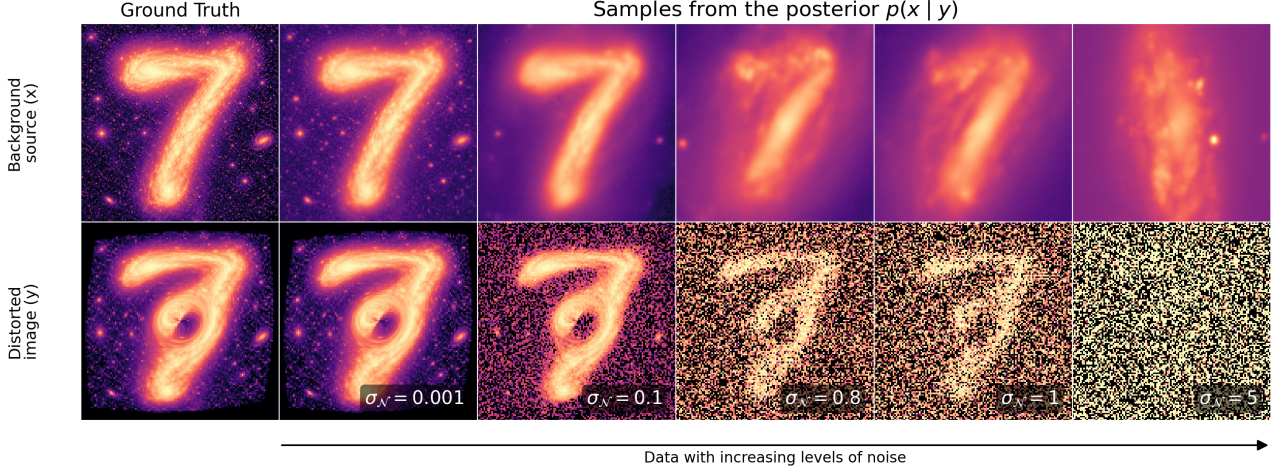


Figure 3: Qualitative OOD reconstruction as observation noise increases. The result shifts from preserving the digit-shaped source to reflecting the learned galaxy prior when the data become less informative.

effectively than drawing additional samples from the same observations. Controlled experiments could test sensitivity to the reverse-process step count and compare a pooling-aware likelihood covariance with the current scalar approximation. Source sets designed to vary flux and morphology separately would help identify which properties are associated with larger residuals. Empirical coverage should also be assessed for interpretable quantities such as flux, size, ellipticity, and concentration.

A matched comparison with conventional reconstruction methods should evaluate accuracy, calibration, computational cost, and downstream parameter recovery under the same simulated conditions. The framework could then be tested under more realistic observational settings, including point-spread functions, detector effects, and lens-model uncertainty.

Research Impact & Conclusion

This project reproduced the principal fixed-lens demonstrations of diffusion-based source reconstruction and established a validation framework that separates implementation fidelity, agreement with the target galaxy distribution, and posterior calibration. The results confirm the central qualitative behaviour of the method, reveal source-dependent residual performance, and identify a discrepancy between the generated and selected PROBES distributions. The posterior diagnostic remained compatible with its matched reference at the available precision.

The main contribution is both an independent implementation and a clearer set of evidence requirements for assessing learned priors before their use in realistic strong lens analyses. Diffusion priors offer a flexible representation of galaxy morphology and potentially complex source uncertainty, but their scientific value should be established through calibration, empirical coverage, and matched comparisons with conventional reconstruction methods.

References

- [1] T. Treu, “Strong Lensing by Galaxies,” *ARA&A* (2010).
- [2] Y. Song et al., “Score-Based Generative Modeling,” *ICLR* (2021).
- [3] A. Adam et al., “Posterior Samples with Score-Based Priors,” *NeurIPS ML4PS* (2022).
- [4] C. Stone et al., “PROBES. I,” *ApJS* (2022).
- [5] P. Lemos et al., “PQMass,” *ICLR* (2025).
- [6] S. Sharief et al., “MIRA,” *arXiv* (2026).